Zhixin MA

(+65) 91418993
zxma@smu.edu.sg
https://zhixinma.github.io

Short Bio

MA Zhixin is currently a Research Scientist at Singapore Management University. He earned his Ph.D. from Singapore Management University in 2024 and his Bachelor's degree from Shandong University in 2019. His research interests span multimedia computing and human-in-the-loop interactive systems. His current work focuses on the grounding and reasoning of MLLMs.

Experience

Research Scientist 2025 - present
Singapore Management University Singapore
- Research area: Cross-Modality Reasoning of MLLMs

Research Assistant 2019 - 2020
National University of Singapore Singapore
- Advisor: Prof. Wenqiang Lei & Prof. Tat-Seng Chua
- Research area: Semantic Parsing

Education

Singapore Management University Singapore
School of Computing and Information Systems
Ph.D. in Computer Science 2020 - 2024
- Advisor: Prof. Chong-Wah Ngo
- Dissertation topic: Interactive Video Search on Large-Scale Corpora

Shandong University Jinan, China
School of Computer Science and Technology
B.E. in Computer Science and Technology (Elite Class) 2015 - 2019
- Advisor: Prof. Tian Gan & Prof. Liqiang Nie

Pre-print

1. **Zhixin Ma***, Yutong Zhou*, Yongqi Li, Chong-Wah Ngo, Wenjie Li. Beyond APIs: Probing the Limits of MLLMs in Physical Tool Use, 2026.

Publications

1. Burak Satar*, **Zhixin Ma***, Cheng Yu-Tong, Huy Hoang Tran, Phuong Anh Nguyen, Chong-Wah Ngo. Cultural Moment Benchmark: Evaluating Video Cultural Reasoning and Grounding in Southeast Asia, *The Conference on Empirical Methods in Natural Language Processing, Main Conference*, 2026.
2. Chao Chen, **Zhixin Ma**, Yongqi Li, Yupeng Hu, Yinwei Wei, Wenjie Li, Liqiang Nie. Reasoning in the dark: Interleaved vision-text reasoning in latent space. *Annual Meeting of the Association for Computational Linguistics, Findings*, 2026.
3. Dongjie Cheng, Yongqi Li, **Zhixin Ma**, Hongru Cai, Wenjie Wang, Liqiang Nie, Wenjie Li. Omni-R1: Towards the Unified Generative Paradigm for Multimodal Reasoning. *Annual Meeting of the Association for Computational Linguistics, Findings*, 2026.
4. Burak Satar*, **Zhixin Ma***, Patrick A. Irawan, Wilfried A. Mulyawan, Jing Jiang, Ee-Peng Lim, Chong-Wah Ngo. Seeing Culture: A Benchmark for Visual Reasoning and Grounding. *The Conference on Empirical Methods in Natural Language Processing, Main Conference*, 2025.
5. **Zhixin Ma**, Chong-Wah Ngo. Robust Relevance Feedback for Interactive Known-Item Video Search. *ACM International Conference on Multimedia Retrieval*, 2025.
6. **Zhixin Ma**, Jiaxin Wu, Chong-Wah Ngo. Leveraging LLMs and Generative Models for Interactive Known-Item Video Search. *International Conference on Multimedia Modeling*, 2024.

Publications

6. **Zhixin Ma**, Chong-Wah Ngo. Interactive Video Corpus Moment Retrieval using Reinforcement Learning. *ACM International Conference on Multimedia*, 2022.
Oral & Top rated paper award: 13 out of 690 accepted papers (1.88%)
7. **Zhixin Ma**, Jiaxin Wu, Zhijian Hou, Chong-Wah Ngo. Reinforcement Learning-Based Interactive Video Search. *International Conference on Multimedia Modeling*, 2022.
8. Wenqiang Lei, Weixin Wang, **Zhixin Ma**, Tian Gan, Wei Lu, Min-Yen Kan and Tat-Seng Chua. Re-examining the Role of Schema Linking Text-to-SQL. *EMNLP*, 2020.

Awards and Honors

- **Presidential Doctoral Fellowship**, Singapore Management University 2023
- **Dean's List**, Singapore Management University 2023
- **The Second Runner-up**, Video Browsing Showdown 2023
- **Top Rated Paper Award**, ACM Multimedia 2022
- **Student Travel Grant**, ACM Multimedia 2022
- **The Second Runner-up**, Video Browsing Showdown 2021

Invited Talk

- **Bridging Cognitive Foundation and Efficient Specialization**
ICMR, Chicago, US 2025
- **Reinforcement Learning Enhanced Interactive Video Retrieval**
NExT++ Research Centre, NUS, Singapore 2023

Community Services

- *Reviewer*: IEEE Transactions on Multimedia (TMM), ACM Transactions on Multimedia Computing, Communications, and Applications (TOMM), IEEE Transactions on Knowledge and Data Engineering (TKDE), IEEE Transactions on Circuits and Systems for Video Technology (TCSVT), ACM Transactions on the Web (TWeb), ACM Transactions on Intelligent Systems and Technology (TIST), ACM International Conference on Multimedia (MM), Annual Meeting of the Association for Computational Linguistics (ACL), Empirical Methods in Natural Language Processing (EMNLP), Annual Conference on Neural Information Processing Systems (NeurIPS), International Joint Conferences on Artificial Intelligence (IJCAI); *Organizer*: Volunteer Chair in ICMR 2027.

Teaching Assistant

- SMT481: Smart-City Operations Research AY2024
- IS446: Managing Customer Relations with Analytics: Asian Insights AY2023
- CS702: Computational Interaction AY2022
- CS601: Introduction to Artificial Intelligence AY2021